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CERN

EDUCATION & OUTREACH DEPARTMENT

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THE CHALLENGE

HIGH-ENERGY COLLISION DATA IS INHERENTLY “NOISY”, CONTAINING MISIDENTIFIED PARTICLES AND BACKGROUND EVENTS.

To use this data for precision physics, we must first quantify detector reliability and establish clear selection criteria to isolate pure physical signals from instrumental "fake" data.

THE OBJECTIVE

01

DISCOVERY

To retrieve the Z-boson resonance from a 100,000-event CMS dataset

02

ANALYSIS

By analyzing electron kinematics (transverse momentum and angular distribution) and charge symmetry

03

VALIDATION

To validate the Standard Model and establish a "Gold Standard" filtered dataset for educational use.

DATASET STRUCTURE

Run	Event	E1	px1	py1	pz1	pt1	eta1	phi1	Q1	E2	px2	py2	pz2	pt2	eta2	phi2	Q2	M
147115	366639895	58.7141	-7.31132	10.531	-57.2974	12.8202	-2.20267	2.17766	1	11.2836	-1.03234	-1.88066	-11.0778	2.14537	-2.34403	-2.07281	-1	8.94841
147115	366704169	6.61188	-4.15213	-0.579855	-5.11278	4.19242	-1.02842	-3.00284	-1	17.1492	-11.7135	5.04474	11.4647	12.7536	0.808077	2.73492	1	15.893
147115	367112316	25.5419	-11.4809	2.04168	22.7246	11.661	1.42048	2.9656	1	15.8203	-1.4728	2.25895	-15.5888	2.69667	-2.45508	2.14857	1	38.3877
147115	366952149	65.3959	7.51214	11.8871	63.8662	14.0619	2.21838	1.00721	1	25.1273	4.08786	2.59641	24.6563	4.84272	2.33021	0.565865	-1	3.72862
147115	366523212	61.4504	2.95284	-14.6227	-59.6121	14.9179	-2.09375	-1.37154	-1	13.8871	-0.277757	-2.4256	-13.6708	2.44145	-2.4237	-1.68481	-1	2.74718

*ONLY 5 ROWS WERE EXTRACTED

DATASET SOURCE

KEY

The ID	**RUN**	Operating period
	EVENT	The specific episode
Energy & Momentum	**E1, E2**	Total particle's speed and mass unleashed energy in the collision (GeV)
	PX, PY, PZ	Detected electron's momentum in the three dimensions (GeV)
	PT1, PT2	How much particles flew away (transverse momentum) (GeV)
Coordinates	**ETA**	Forward/backward angle (pseudorapidity)
	PHI	Clock face angle (rad)
Charge	**Q1, Q2**	Electric charge of the electron (-1 = standard electron, +1 = positron. Always complementary)
The result	**M**	Invariant mass of two electrons (GeV) (Useful to calculate the instantly vanished parent particle mass)

QUESTIONS

GOOGLE COLAB NOTEBOOK

CHARGE ANALYSIS

HOW MANY ELECTRONS HAVE CHARGE +1 VS -1 IN Q1 AND IN Q2?

charge	Q1	Q2
-1	50274	50211
1	49726	49789

FAKE DATA ANALYSIS

HOW MUCH FAKE (SAME SIGN) DATA IS PRESENT IN THE DATASET?

n_tot	n_same_sign	n_opposite_sign	same_sign_rate	opposite_sign_rate
100000	42947	57053	0.42947	0.57053

RETURN THE RUN THAT HAS THE HIGHEST NUMBER OF EVENTS IN THE 80-100 GEV Z-BOSON MASS WINDOW

HEAVY Z-BOSON MASS

Run	n_Z
149181	1981

ENERGY & MOMENTUM ANALYSIS

WITHIN THE Z-MASS WINDOW, CONSIDER THE MOST VIOLENT EVENTS, WHERE AT LEAST ONE ELECTRON PT > 50 GEV. FOR EACH EVENT TAKE THE HIGHER OF THE TWO ELECTRON PT VALUES. THEN REPORT THE SMALLEST VALUE AMONG THIS EVENTS.

50.0163 GEV

FOR EACH RUN, COMPARE THE MOST VIOLENT EVENTS INSIDE AND OUTSIDE THE Z-MASS WINDOW.

REPORT THE EVENT COUNTS AND THE MEAN LEADING AND SUBLEADING ELECTRON PT IN EACH REGION.

n_in	pt_lead_in_mean	pt_sublead_in_mean	Run	n_out	pt_lead_out_mean	pt_sublead_out_mean
34	68.46	34.56	146511	24	60.47	6.83
80	60.35	33.13	146644	111	62.81	7.98
45	66.04	32.72	146944	47	61.08	9.54
55	62.15	34.95	147114	59	63.38	7.41
39	65.62	32.61	147115	43	66.75	9.01
55	64.16	33.62	147754	81	65.31	7.86
73	62.87	34.67	147926	59	67.03	10.48
66	61.81	35.55	147929	73	64.11	10.72
68	65.53	32.89	148029	75	63.46	10.62
77	64.82	34.72	148031	88	66.33	9.52
206	61.89	33.96	149011	189	63.63	9.33
390	61.57	34.46	149181	371	65.52	9.66
66	61.27	33.54	149182	75	64.83	8.82

BACK-TO-BACK ANALYSIS

QUANTIFY HOW CLOSE ELECTRON PAIRS ARE TO BEING BACK-TO-BACK IN Φ AND HOW BALANCED THEY ARE IN TRANSVERSE MOMENTUM

avg_nearly_back_to_back	avg_acoplanarity	avg_pt_imbalance_ratio
0,108	1.199	0,533

WHAT FRACTION OF ALL EVENTS ARE NEARLY BACK-TO-BACK IN Φ AND BALANCED IN PT?

0.047 GEV

ELECTRON PAIR CATEGORY

WHAT IS THE OPPOSITE CHARGE FRACTION FOR EACH "ELECTRON PAIR CATEGORY" AND NUMBER OF EVENTS?

eta_pair	n_events	opposite_sign_rate
Central-Forward	42652	0,554
Forward-Forward	36298	0,536
Central-Central	18441	0,683
Out of Acceptance	2609	0,531

CONCLUSIONS

GIVEN THE NUMBERS OF POSITIVE AND NEGATIVE CHARGES, THE DATASET DOES NOT APPEAR TO BE CHARGE-IMBALANCED. HOWEVER, THERE IS A LARGE AMOUNT OF NOISE TO REMOVE, DUE TO A SIGNIFICANT CONTAMINATION OF SAME-SIGN ELECTRON PAIRS.

WHEN WE RESTRICT THE ANALYSIS TO EVENTS WITHIN THE Z MASS WINDOW (80–100 GEV), THE OPPOSITE-SIGN FRACTION INCREASES SHARPLY, SHOWING THAT THE INVARIANT MASS IS A DISCRIMINATING VARIABLE FOR SELECTING A MUCH CLEANER SUB-POPULATION. EVEN AFTER THIS SELECTION, THE DATASET IS NOT UNIFORM ACROSS RUNS, WITH SOME RUNS CONTRIBUTING MANY MORE EVENTS THAN OTHERS.

CONCLUSIONS

COMPARING THE MOST VIOLENT COLLISIONS INSIDE AND OUTSIDE THE Z MASS WINDOW, WE FIND THAT THE LEADING PT IS SIMILAR IN BOTH REGIONS, WHILE THE SUBLEADING PT IS MUCH HIGHER INSIDE THE WINDOW. THIS SUGGESTS THAT OUTSIDE THE WINDOW ONE ELECTRON TENDS TO BE VERY ENERGETIC, WHILE THE OTHER IS MUCH SOFTER.

OVERALL, THE DATASET SHOWS A LOW FRACTION (ABOUT 4%) OF EVENTS IN WHICH THE ELECTRON PAIR IS NEARLY BACK-TO-BACK AND BALANCED IN TRANSVERSE MOMENTUM. FINALLY, THE ELECTRON-PAIR CATEGORY IN |ETA| MATTERS FOR PURITY: THE CENTRAL–CENTRAL CATEGORY, EVEN THOUGH IT CONTAINS FEWER EVENTS, HAS THE HIGHEST OPPOSITE-CHARGE FRACTION.

THANK YOU